

Full Stack Development with MERN

Project Documentation format

1. Introduction

Project Title: ShopEZ – E-Commerce Platform

Team Members: Pothana Sreehitha

2. Project Overview

Purpose: ShopEZ is a comprehensive MERN stack-based e-commerce platform designed for exploring products, managing a shopping cart, secure checkouts, and providing advanced administrative controls for inventory and order management.

Features: User Registration, Login (Customer & Admin), JWT Authentication, Product Catalog & Search, Shopping Cart Management, Order Placement & Tracking, Dedicated Admin Dashboard (Users, Products, Orders), and Responsive UI.

3. Architecture

Frontend: React.js, React Router, Axios, Custom CSS.

Backend: Node.js, Express.js, JWT Authentication, Bcrypt password hashing.

Database: MongoDB with collections for Users, Products, Orders, and Carts.

4. Setup Instructions

Prerequisites: Node.js, MongoDB, npm.

Backend: cd backend -> npm install -> npm run dev

Frontend: cd frontend -> npm install -> npm run dev

5. Folder Structure

Frontend: src, pages, components, context.

Backend: controllers, models, routes, middleware, config, data.

6. Running the Application

Backend: npm run dev

Frontend: npm run dev

7. API Documentation

POST /api/users/register

POST /api/users/login

GET /api/users/profile

GET /api/products

GET /api/products/:id

GET /api/cart

POST /api/cart

POST /api/orders

GET /api/orders/:id

PUT /api/orders/:id/status

8. Authentication

JWT-based authentication combined with bcrypt password hashing. Includes protected routes with role-based access control (Customer vs. Admin).

9. User Interface

Pages: Home, Login, Register, Product Details, Cart, Place Order, Order Status, Customer Profile, Admin Dashboard.

10. Testing

Tools Used: Postman / Thunder Client, MongoDB Compass, Browser Developer Tools.

11. Screenshots or Demo

GitHub Repository: <https://github.com/sreehithapothana/shopez-e-commerce>

12. Known Issues

Payment processing is currently simulated for demonstration purposes.

13. Future Enhancements

Integration with Stripe/Razorpay for real payments, Product Reviews & Ratings, Real-time Inventory Alerts, and Automated Email Notifications.